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Pedro Miguel Sánchez Sánchez

Data Scientist and Researcher — AI, Federated Learning, Cybersecurity & Recommender Systems

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About Me

Data Scientist building production recommender systems at Roche, where I own personalisation for global Diagnostics sales teams. PhD Cum Laude (University of Murcia, armasuisse S&T collaboration) on ML-based detection of compromised and spoofed devices, including adversarial attacks and defences against detection models. I followed as a postdoc on distributed and trustworthy AI, spanning Federated Learning, security, and model explainability. National Young Researcher Award in Computer Science, 2024.

Experience

Data Scientist, *Roche* *Madrid, Spain Sept 2025 – Present*

- Designed and deployed an AI recommender system for global Roche Diagnostics sales teams, directly supporting new sales opportunities and revenue growth.
- Built the full modelling loop in production: candidate generation, ranking, feature pipelines over customer, product and sales data, plus versioning, monitoring and scheduled retraining.

Senior Researcher, *Funditec (Advantx Technology Foundation)* *Madrid, Spain Nov 2024 – Sept 2025*

- Led ML/DL and cybersecurity research across European and National projects, contributing to proposal writing and technical execution from grant acquisition through delivery.

Postdoctoral Researcher, *University of Murcia* *Murcia, Spain Feb 2024 – Nov 2024*

- Developed novel Federated Learning methods and trustworthy AI frameworks in direct collaboration with armasuisse S&T (Switzerland), worked in AI robustness, privacy, explainability and fairness.
- Drove research contributions to two funded EU projects: EU-GUARDIAN (European Defence Fund) and ROBUST-6G (Horizon Europe).

Visiting Researcher, *Cyber-Defence Campus, armasuisse S&T* *Thun, Switzerland Sept–Dec 2021, 2022*

- Conducted two research residencies (6 months total) at Switzerland's Cyber-Defence Campus, producing results that directly fed into PhD thesis publications and an ongoing armasuisse collaboration.

Predoctoral Researcher, *University of Murcia* *Murcia, Spain Sept 2019 – Feb 2024*

- Delivered armasuisse S&T-funded research on IoT device fingerprinting and adversarial attack using ML/DL, culminating in a Cum Laude PhD, a national award, and 12+ peer-reviewed publications.

Education

Ph.D. in Computer Science, Cum Laude. *University of Murcia* *Sept 2019 – Feb 2024*

- Thesis: *Identical IoT device identification via hardware performance fingerprinting and Machine Learning*
- Supported by *armasuisse S&T*, also involved in the EU-GUARDIAN EDF and 5GZORRO H2020 projects

M.Sc. in Computer Science, 9.07/10. *University of Murcia* *Sept 2018 – June 2019*

- Thesis: *Securing Smart Offices through an Intelligent and Multi-device Continuous Authentication System*

B.Sc. in Computer Science, 8.04/10. *University of Murcia* *Sept 2014 – June 2018*

- Graduated with honors. Networking Specialization. Thesis: *Continuous authentication for mobile devices*

Research

21 h-index (+20 JCR journals, +10 confs, +4 chapters). Relevant research on Federated Learning, applied ML/DL in IoT behavior security, and AI Trustworthiness. Updated publication list in Google Scholar.

- Contributor to NEBULA, an open source platform for decentralized model training, and to the EU-GUARDIAN (EDF), ROBUST-6G and 5G-ZORRO (Horizon) projects.

Awards and Achievements

National Young Researcher Award in Computer Science *2024*

- Granted by the Scientific Society of Computer Science of Spain and the BBVA Foundation

Outstanding PhD Thesis Award *2024*

- Award for the best Computer Science PhD Thesis in 2024 at my university (University of Murcia)

Elsevier Computer Networks Best Paper Award *2024*

- "Federated learning for malware detection in IoT devices", a collab with EPFL, UZH, and CYD Campus

Technical Skills

Skills: Federated Learning, Recommender Systems, AI Trustworthiness, Machine/Deep Learning, Security, IoT

Programming and Frameworks: Python, Bash, Keras, PyTorch, scikit-learn, C/C++, AWS SageMaker, Docker, Git

Annex. Main Publications

[1] **Sánchez Sánchez, Pedro Miguel**, E. T. Martínez Beltrán, M. Fernández Llamas, G. Bovet, G. Martínez Pérez, and A. Huertas Celdrán, "Profe: Communication-efficient decentralized federated learning via distillation and prototypes," in *IEEE International Conference on Communications*, 2025, pp. 1596–1601.

[2] E. T. Martínez Beltrán, **Sánchez Sánchez, Pedro Miguel**, G. Bovet, B. Stiller, G. Martínez Pérez, and A. Huertas Celdrán, "Flighter: Decentralized federated learning and situational awareness for secure military aerial reconnaissance," *IEEE Communications Magazine*, vol. 63, no. 10, pp. 136–142, 2025.

[3] L. C. Seller, Í. S. Torres, A. Vogel-Fernández, C. G. Carballo, **Sánchez, Pedro Miguel Sánchez**, A. C. Martín, and E. d. M. Ambite, "Evaluating compact llms for zero-shot iberian language tasks on end-user devices," *SEPLN: Procesamiento del lenguaje natural 75*, 2025.

[4] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, G. Bovet, and G. Martínez Pérez, "Transfer learning in pre-trained large language models for malware detection based on system calls," in *MILCOM 2024-2024 IEEE Military Communications Conference (MILCOM)*. IEEE, 2024, pp. 853–858.

[5] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, N. Xie, G. Bovet, G. Martínez Pérez, and B. Stiller, "Federatedtrust: A solution for trustworthy federated learning," *Future Generation Computer Systems*, vol. 152, pp. 83–98, 2024.

[6] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, G. Bovet, and G. Martínez Pérez, "Adversarial attacks and defenses on ml-and hardware-based iot device fingerprinting and identification," *Future Generation Computer Systems*, vol. 152, pp. 30–42, 2024.

[7] A. Huertas Celdrán, **Sánchez Sánchez, Pedro Miguel**, J. Von Der Assen, T. Schenk, G. Bovet, G. Martínez Pérez, and B. Stiller, "Rl and fingerprinting to select moving target defense mechanisms for zero-day attacks in iot," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 5520–5529, 2024.

[8] C. Feng, A. Huertas Celdrán, J. Baltensperger, E. T. Martínez Beltrán, **Sánchez Sánchez, Pedro Miguel**, G. Bovet, and B. Stiller, "Sentinel: An aggregation function to secure decentralized federated learning," in *European Conference on Artificial Intelligence (ECAI) 2024*. IOS Press, 2024, pp. 1760–1767.

[9] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, G. Bovet, and G. Martínez Pérez, "Single-board device individual authentication based on hardware performance and autoencoder transformer models," *Computers & Security*, vol. 137, p. 103596, 2024.

[10] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, T. Schenk, A. L. B. Iten, G. Bovet, G. Martínez Pérez, and B. Stiller, "Studying the robustness of anti-adversarial federated learning models detecting cyberattacks in iot spectrum sensors," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 2, pp. 573–584, 2024.

[11] E. T. Martínez Beltrán, **Sánchez Sánchez, Pedro Miguel**, S. López Bernal, G. Bovet, M. Pérez Gil, G. Martínez Pérez, and A. Huertas Celdrán, "Fedstellar: A platform for training models in a privacy-preserving and decentralized fashion," in *International Joint Conference on Artificial Intelligence (IJCAI-23) Demos*, 2023.

[12] A. Huertas Celdrán, J. Kreischer, M. Demirci, J. Leupp, **Sánchez Sánchez, Pedro Miguel**, M. Franco Figueredo, G. Bovet, G. Martínez Pérez, and B. Stiller, "A framework quantifying trustworthiness of supervised machine and deep learning models," in *SafeAI2023: The AAAI's Workshop on Artificial Intelligence Safety*, 2023, pp. 2938–2948.

[13] E. T. Martínez Beltrán, M. Pérez Quiles, **Sánchez Sánchez, Pedro Miguel**, S. López Bernal, G. Bovet, M. Pérez Gil, G. Martínez Pérez, and A. Huertas Celdrán, "Decentralized federated learning: Fundamentals, state of the art, frameworks, trends, and challenges," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 4, p. 2983–3013, 2023.

[14] V. Rey, **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, and G. Bovet, "Federated learning for malware detection in iot devices," *Computer Networks*, vol. 204, p. 108693, 2022.

[15] **Sánchez Sánchez, Pedro Miguel**, J. M. Jorquera Valero, A. Huertas Celdrán, G. Bovet, M. Pérez Gil, and G. Martínez Pérez, "A survey on device behavior fingerprinting: Data sources, techniques, application scenarios, and datasets," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 2, pp. 1048–1077, 2021.